

Rotary

A two-rotor rotary speaker cabinet: valve drive, treble horn and bass drum, two mics.



Overview

Rotary is a rotary speaker cabinet in the spirit of the classic organ cabinets. The signal first goes through a valve amp (DRIVE), for the growl when you dig in. A crossover at 800 Hz then splits it between a treble HORN and a bass DRUM, which spin separately, each with its own speeds and its own weight: the horn gets up to speed in well under a second, while the heavy drum takes seconds. That difference is what makes the speed change sound so alive.

As each rotor spins you hear the pitch move (Doppler), the level swell, and - for the horn - the tone brighten and darken as it points towards and away from the mics, plus its reflection off the cabinet. Two mics pick it up: DISTANCE trades swirl for cabinet sound, and SPREAD goes from mono to wide stereo. The speed can be switched from the panel, from automation, or from MIDI on the track (mod wheel or sustain pedal), just like the half-moon switch and foot switch of a real organ.

The cabinet takes a mono signal: the two inputs are added together before the amp.

Reach for it for

- Organ, of course - drawbar and combo organs come alive
- Electric piano and clav for a swirling, vintage sound

- Guitar through a rotary cabinet
- Vocals or pads for a woozy, psychedelic effect
- Speed-change swells (CHORALE to TREMOLO) as a performance effect

Controls

The window

A view from above onto the HORN and DRUM as they spin, with each rotor's current speed in Hz. The two amber dots show where the mics sit (set by SPREAD). The line underneath says what the cabinet is doing: CHORALE (slow), TREMOLO (fast), SPEEDING UP, SLOWING DOWN, or BRAKE when stopped.

ROTOR SPEED

SLOW / FAST (half-moon switch)	Click the left half for SLOW (chorale) or the right half for FAST (tremolo). The rotors speed up or slow down to the new setting at their own pace.
STOP	The brake: both rotors coast to a stop. Click it again to start them at the last speed you used.

The rotors

HORN SLOW	The horn's slow speed, 0.3 to 1.5 Hz (0.83 Hz by default).
HORN FAST	The horn's fast speed, 4 to 8 Hz (6.75 Hz by default).
HORN ACCEL	How long the horn takes to change speed, 0.2 to 3 s (0.6 s by default). Slowing down takes a little longer than speeding up.
DRUM SLOW	The drum's slow speed, 0.2 to 1.5 Hz (0.67 Hz by default).
DRUM FAST	The drum's fast speed, 3.5 to 7.5 Hz (5.8 Hz by default).
DRUM ACCEL	How long the heavy drum takes to change speed, 1 to 10 s (4.5 s by default).

MIDI SPEED

Rotary switches speed from MIDI on its own track, and passes all MIDI on. If the instrument in front of it doesn't pass MIDI through, set that instrument's MIDI output to 'merge with input'. MIDI does nothing while STOP is on.

MOD WHL	On by default. Mod wheel up (past halfway) = FAST, down = SLOW.
PEDAL	On by default. Each press of the sustain pedal toggles between SLOW and FAST.

CABINET

DRIVE	The valve amp in front of the cabinet, 0 to 100% (20% by default). Adds warmth, then growl and grit as you turn it up, with a slightly darker tone. The level stays roughly steady.
BALANCE	Horn against drum, -6 to +6 dB. Positive gives more horn (brighter, more swirl on top); negative gives more drum.
DISTANCE	How far the mics are from the cabinet, 0 to 100% (50% by default). Close mics hear more of the swell and swirl; far mics hear more of the cabinet and the horn's reflection.
SPREAD	The angle between the two mics, 0 to 100% (70% by default). 0 is mono; higher settings give a wider stereo image, up to the mics on opposite sides.

Drift

MOTOR DRIFT: the motors wander slightly in speed, like a well-used cabinet.

DRIFT	How much drift, 0 to 100%. 50% (the default) is the standard amount, 100% is twice that, 0 switches it off. More drift also makes the wander a little faster.
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MASTER

MIX	Dry / cabinet balance. 100% (the default) is the cabinet only; lower settings blend the dry signal back in.
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OUTPUT	Output level, -24 to +12 dB.
OUT meters	Output level, left and right.

IN / OUT

TRIM	Input level before anything else, from -24 to +12 dB.
GUARD	Click the TRIM knob (without moving it) to switch the guard; its light sits at the knob's upper right and is lit while it is on. On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. The light glows brighter while it is catching peaks.
SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.

The header

PRESETS	Opens the list of factory presets (a click on the display does the same). Choosing one sets the rotor speeds and the cabinet; the rotor speed switch, MIDI keys, drift, master and the IN / OUT section are left as they are. The last entry resets everything to the defaults.
◀ / ▶	Step to the previous or next preset.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Next to the S / M / L size keys. Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Presets

Classic Cabinet	The standard speeds and acceleration, 20% drive, mics at medium distance and spread.
Soul Jazz Trio	Slightly faster fast speeds and quicker acceleration, 40% drive, a touch more horn, closer mics.
Modern Jazz Combo	Slower speeds, gentle 10% drive, mics further back and wider.
Gospel Scream	Fast speeds and quick acceleration, 75% drive, more horn and close mics.
Memphis Groove	Standard speeds, 30% drive, a little more drum, narrower mics.
Slow Burn	Slow rotors that take their time to change speed (1.4 s horn, 7.5 s drum).
Guitar Through A Rotary	A faster slow speed, 35% drive and +4 dB of horn, for guitar.
Close Mics	The classic cabinet with the mics right up close and wide: maximum swirl.
Room Mics	The classic cabinet heard from across the room, with narrower mics.
Reset all to defaults	Every control back to its starting value.

Tips

- The magic is in the speed change: switch between SLOW and FAST in the music, and let the drum lag behind the horn.
- Map the mod wheel or sustain pedal (MOD WHL / PEDAL) so you can switch speed while you play, as on a real organ.
- For grit on organ, push DRIVE to 50% or more and play harder chords.
- If the swirl is too much in a mix, raise DISTANCE or lower SPREAD rather than turning MIX down.

File: gfx-rotary.jsfx